

NEMA 23 Pan/Tilt Gimbal — Assembly Guide

For fabrication / professional assembly. This guide covers the mechanical pan/tilt gimbal only (not the vision or trigger electronics). It is a **design-intent package**: standard interface dimensions are given exactly; build-specific frame dimensions appear as labeled callouts (**A–J**) in the [Dimension Schedule](#) and must be finalized against the supplied parts/scans before machining. Drawings are schematic and not to scale unless a dimension is shown.

Scope boundary: the gimbal ends at the **tilt-output interface plate** (PA-4) — a generic, payload-agnostic mount. Everything above it (marker, cameras, lidar, Pi, trigger) is a separate 3D-printed **payload module** (see `Payload_Top_Module.pdf`). Because the gimbal is payload-agnostic, it needs **no part scans** — it is built entirely from the load spec (≤ 10 lb, CG ≤ 100 mm from the tilt axis, within the bounding envelope).

Revision: C — drivetrain confirmed (motor + gearbox part numbers locked). Companion files: `Gimbal_Build_Plan_Illustrated.pdf`, `Turret_Full_Parts_List.xlsx`.

1. Bill of materials — mechanical

Item	Spec	Qty	Notes
Motor + driver	NEMA 23 closed-loop 3.0 N·m + CL57T V4.1 driver — STEPPERONLINE kit, PN 23HS45-4204D-E1000 (Amazon B0C6943QBM)	2	Pan + tilt; Ø8 shaft, 1000-line encoder
Planetary gearbox	5:1, Ø8 input bore , 15 arcmin; output Ø14 h7 shaft (key 5, M5×12) / Ø40 h7 register / 4×Ø5.2 @ 47.14	2	Matches motor Ø8 shaft
Pan bearing	Slewing/thrust bearing, bore per C	1	Carries payload + moment load
Tilt shaft	Steel, Ø E (e.g. 12 mm), length per D	1	Hardened/ground preferred
Tilt bearings	Flanged or pillow-block, bore to suit E	2	Slip fit on shaft
Shaft collars	Clamp type, Ø E	2–4	Axial location
Reduction coupling	Rigid/bellows coupling, or GT3 belt + 5:1 pulleys	per axis	If belt: idler for tension
Frame stock	Aluminum — 40×40 T-slot extrusion or 6 mm plate/channel	per A,B,W	Metal in the load path
Pan turntable plate	Aluminum, machined	1	Mounts to pan bearing
Tilt yoke	Aluminum plate/bracket	1–2	Carries tilt bearings
Tilt-output interface plate	Aluminum, machined — Ø70 bolt circle (4×M5) + Ø30 h7 pilot + 2×Ø4 dowels	1	Generic payload mount (PA-4); payload-agnostic
Tripod adapter	Plate + anti-rotation key/pins	1	Per PA-5
Fasteners	M5 SHCS (motors), M6 (frame/tripod), M3–M4 (components)	assorted	Class 8.8 min
Dowel pins	Ø3–4 mm, alignment	as req.	Locate critical joints
Consumables	Medium threadlocker, anti-seize, shim stock	—	—

Drive electronics (for reference / functional test — not part of the mechanical build): the **CL57T drivers ship with the motor kits**; you still need 1 × Pico RP2040 motion coprocessor and 1 × 24–48 V supply. See the main turret BOM.

2. Tools & consumables

Metric hex/Allen set · torque wrench (1–10 N·m range) · vernier/digital calipers · dial indicator + magnetic base · combination square · digital angle gauge or precision level · dead-blow mallet · bearing press or arbor (if any press fit) · tap & die set (if tapping frame) · drill + deburring tools · belt tension gauge (if belt reduction) · medium threadlocker · anti-seize.

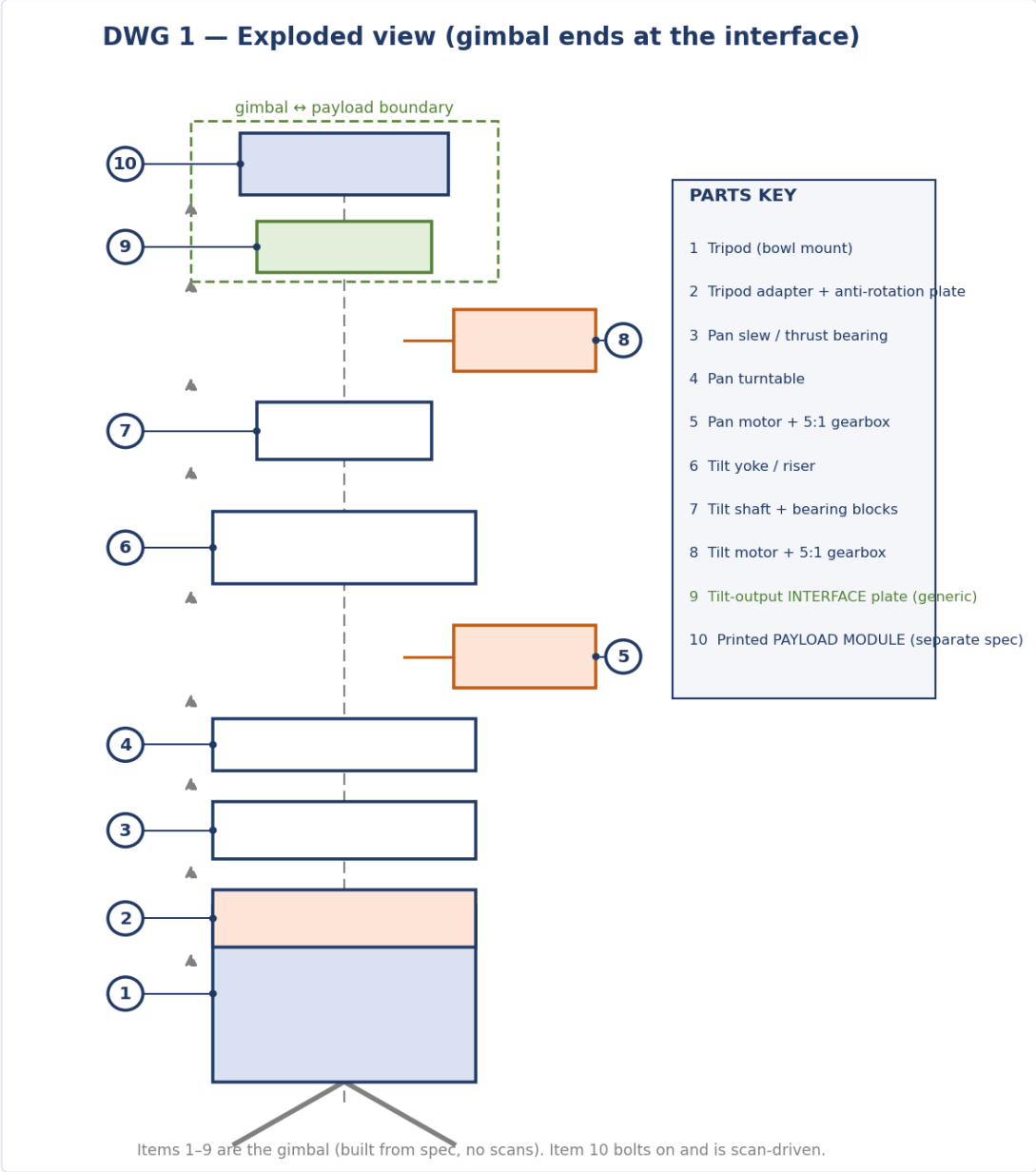
3. Fastener & torque schedule

Class 8.8 steel into aluminum, dry/threadlocked — **confirm against your joint and any printed parts (reduce for plastic):**

Thread	Torque	Typical use
M3	1.0–1.3 N·m	Components, lidar
M4	2.5–3.0 N·m	Brackets, plate
M5	5.0–6.0 N·m	Motor/gearbox mounts
M6	8.5–10 N·m	Frame, tripod interface

Use washers under SHCS heads on aluminum; medium threadlocker on all dynamic/vibrating joints; do **not** thread metal fasteners directly into printed plastic — use heat-set inserts.

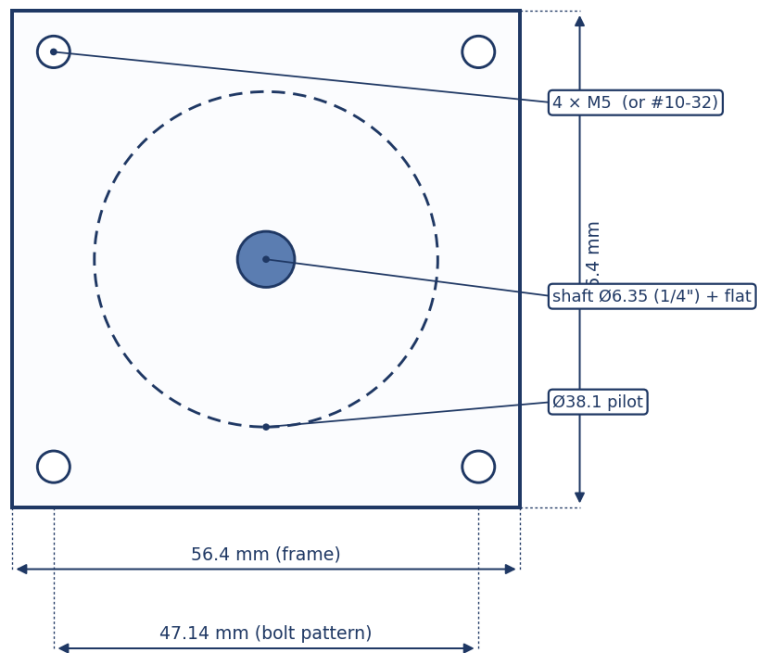
4. Overview — exploded assembly



Assembly order runs bottom → top. Items 5 and 8 (motor + gearbox) mount off-axis and couple in through the reduction.

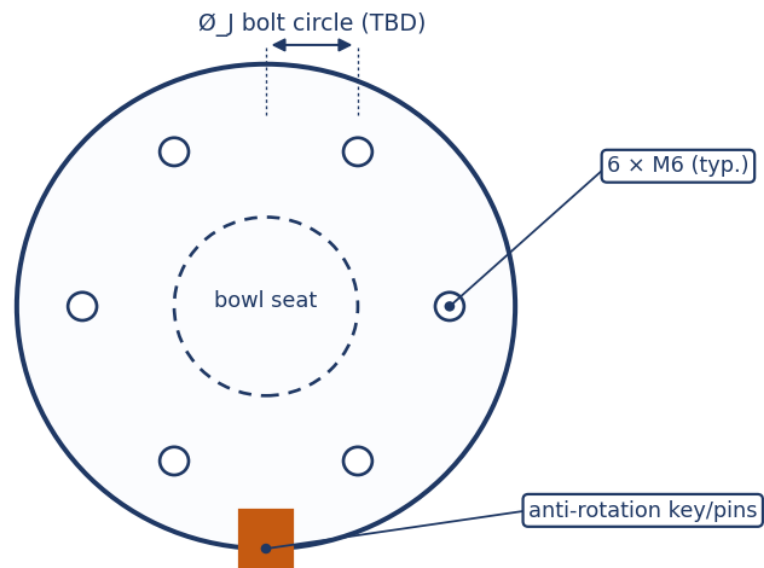
5. Reference interfaces (standard dimensions)

PA-1 NEMA 23 mounting interface (standard — confirm vs motor/gearbox datasheet)



Tip: the gearbox input flange matches this pattern; its OUTPUT flange/shaft is per the chosen gearbox.

PA-5 Tripod interface (pan base, bottom)



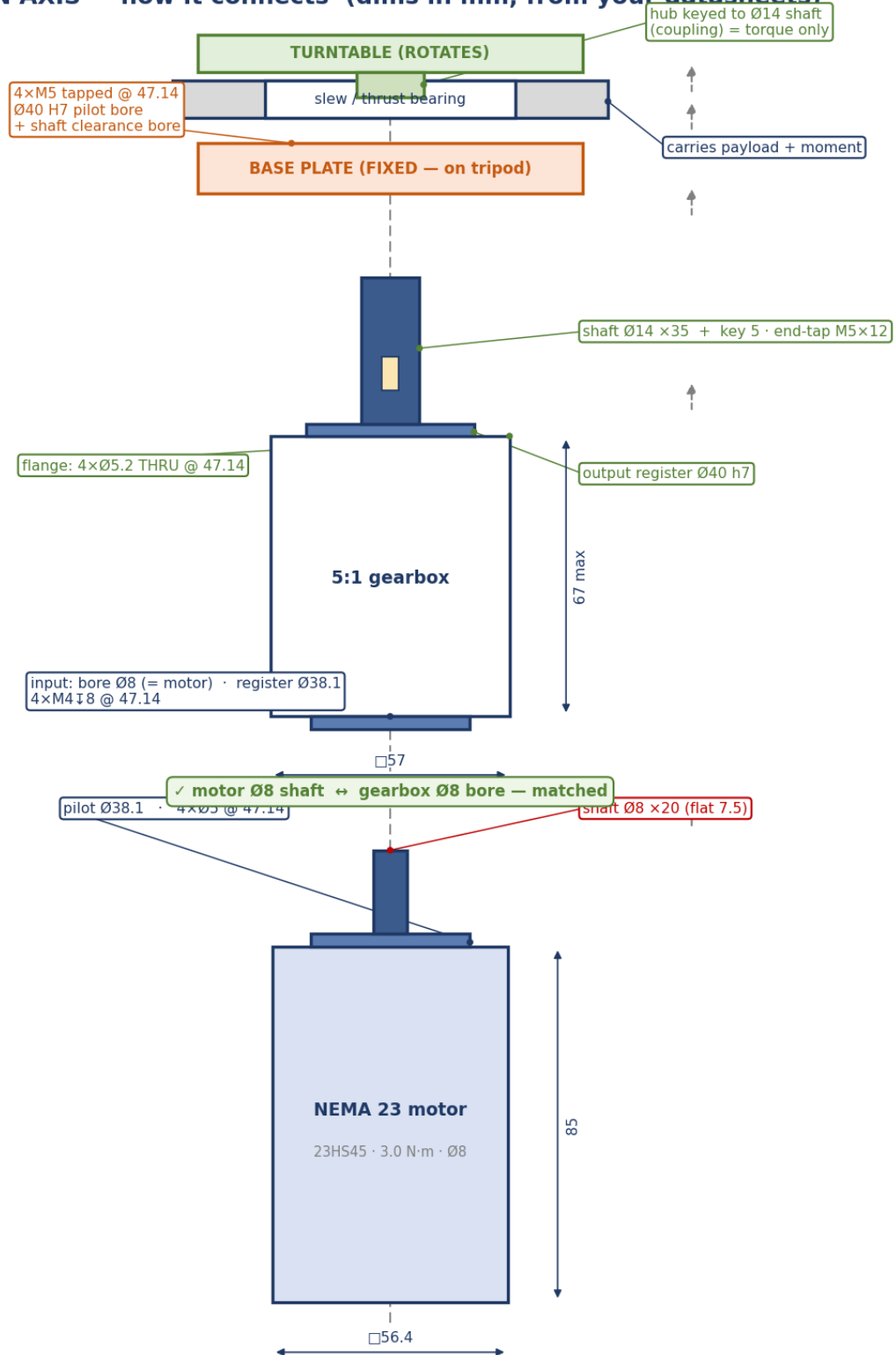
Bolt circle + key resist pan reaction — never a single 3/8" stud.

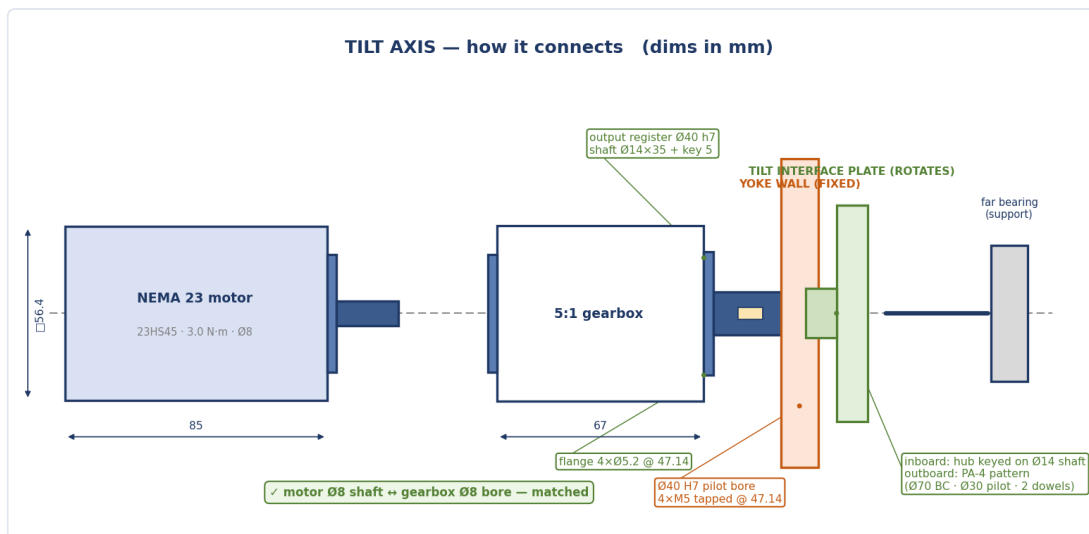
The NEMA 23 pattern (56.4 mm frame, 47.14 mm bolt pattern, Ø38.1 mm pilot, 4 × M5) is standard for both motor and gearbox input — **confirm against the datasheets of the exact units purchased**. The tripod interface uses a bolt circle plus an anti-rotation key.

5a. Confirmed drivetrain (motor + gearbox)

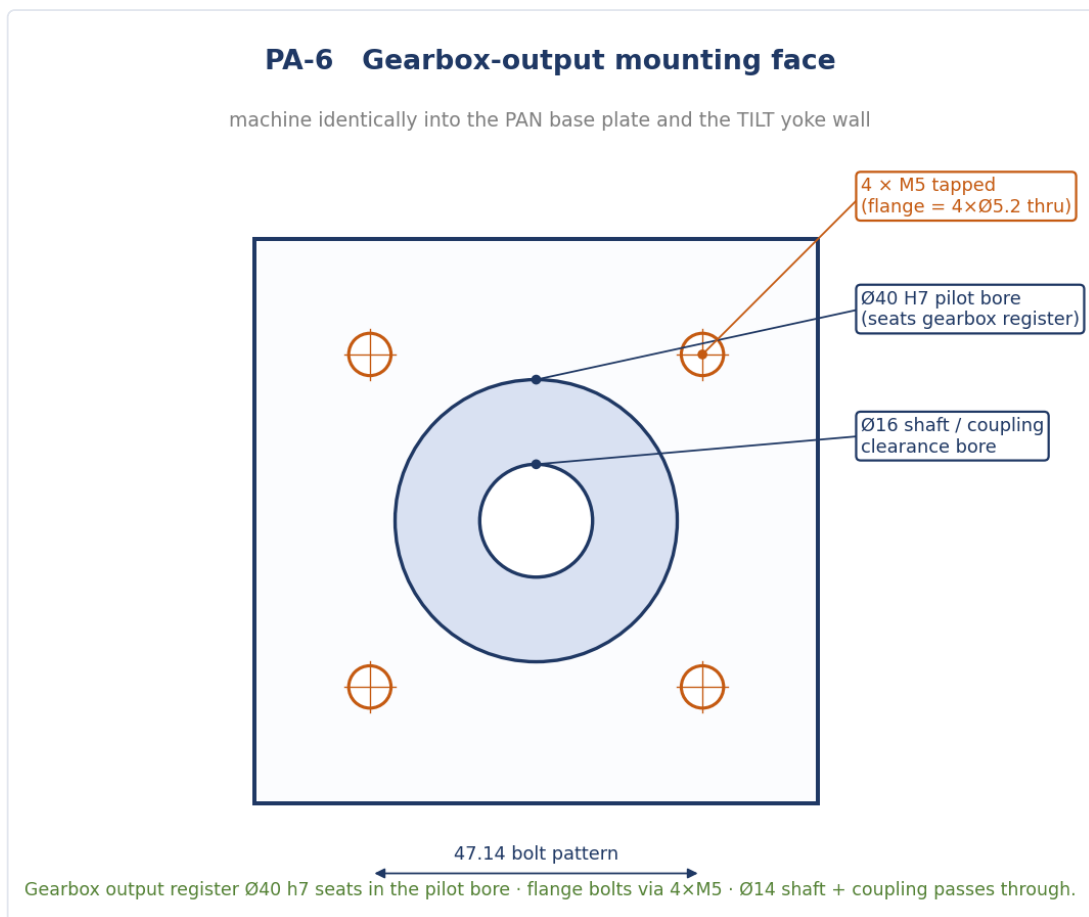
Both axes use the same matched pair: a **NEMA 23 closed-loop 3.0 N·m motor (PN 23HS45-4204D-E1000) + CL57T driver** kit driving a **5:1 planetary gearbox with an Ø8 input bore** (15 arcmin backlash). The Ø8 motor shaft matches the Ø8 gearbox bore, and the Ø38.1 pilot + 47.14 pattern mate directly. Torque at each axis $\approx 3.0 \times 5 = 15 \text{ N·m}$.

PAN AXIS — how it connects (dims in mm, from your datasheets)



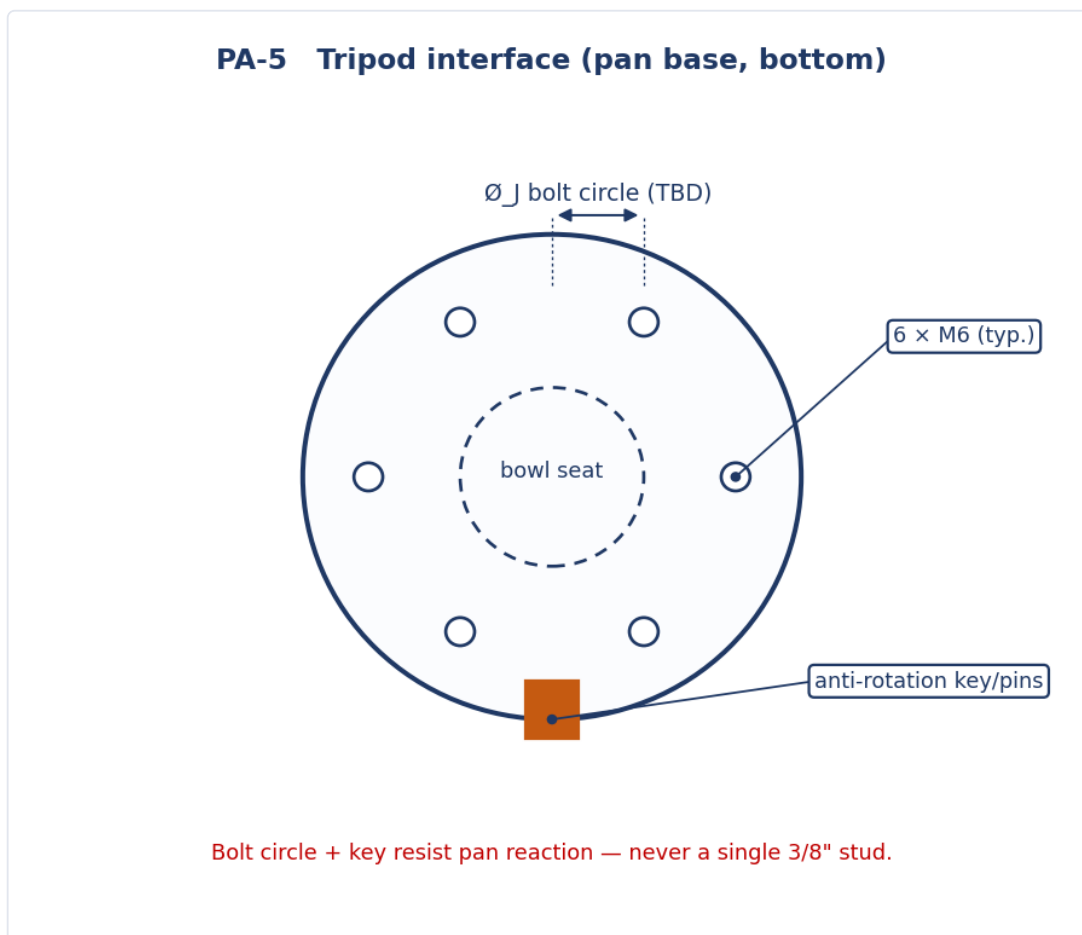


The gearbox **output** is identical on both axes — Ø14 h7 shaft (5 mm key, M5×12 end-tap), Ø40 h7 register, 4×Ø5.2 flange holes on 47.14 — so the face machined into the pan base plate and the tilt yoke wall is the same:



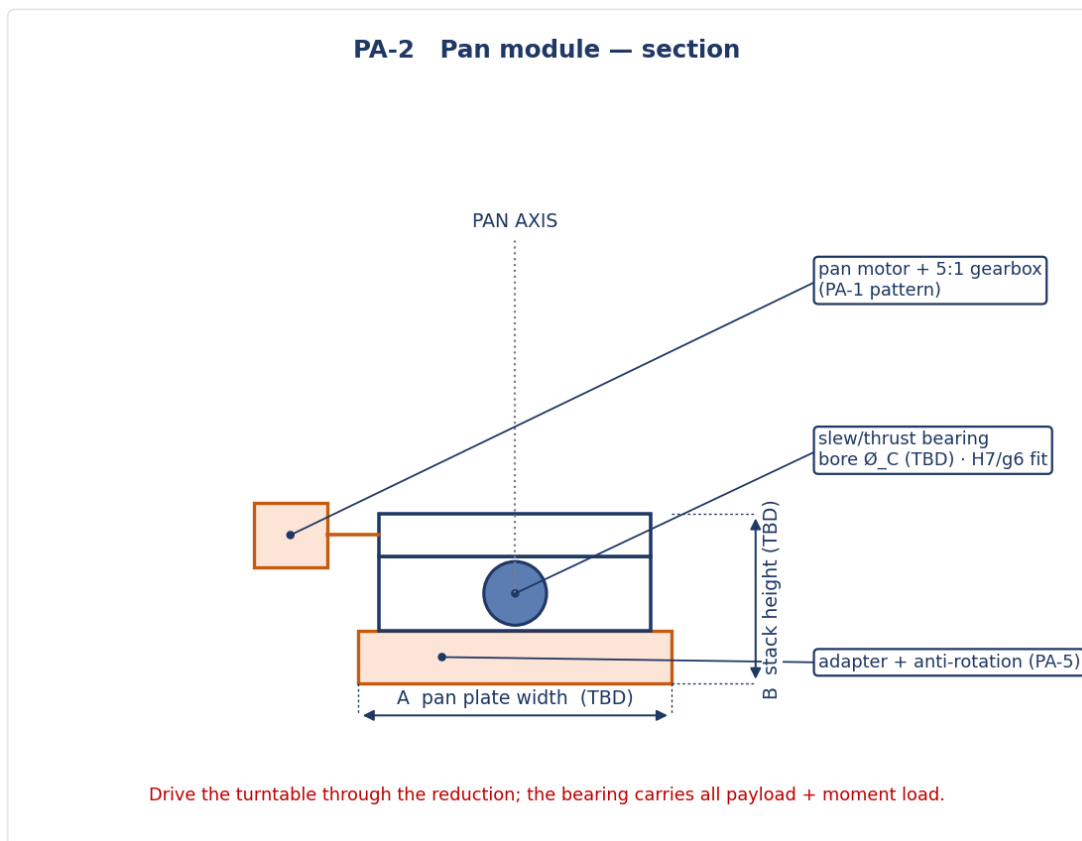
6. Assembly procedure

Step 1 — Tripod adapter & pan bearing



Fit the anti-rotation key/pins and mount the adapter to the pan bearing's lower race. Bolt circle per **J**, M6 @ 8.5–10 N·m, threadlocked. **QC**: adapter face square to the bearing axis (combination square); the bearing rotates freely with no axial play.

Step 2 — Pan motor, gearbox & turntable

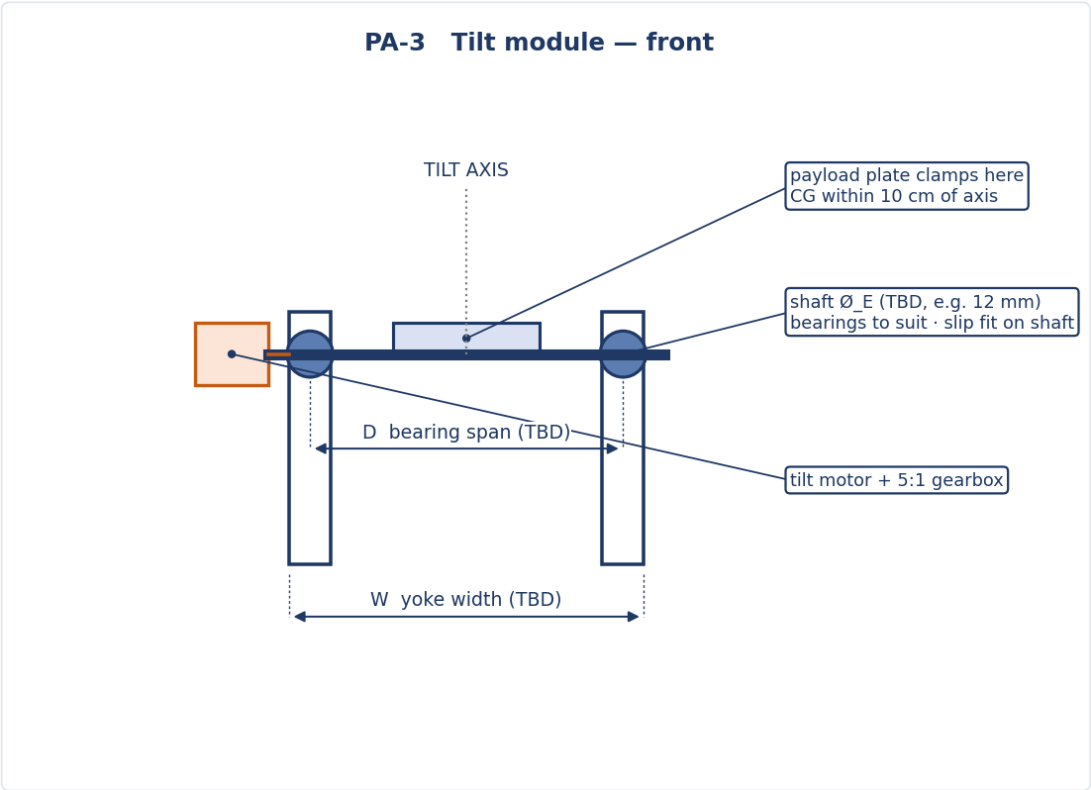


Bolt the 5:1 gearbox to the pan motor (PA-1 pattern, M5 @ 5–6 N·m). Mount the assembly to the frame and couple the output to the turntable through the reduction — **the bearing carries the load, never the motor shaft**. Mount the turntable to the bearing upper race. **QC:** turntable runout ≤ 0.05 mm TIR (dial indicator); free rotation, even torque through 360°.

Step 3 — Tilt yoke

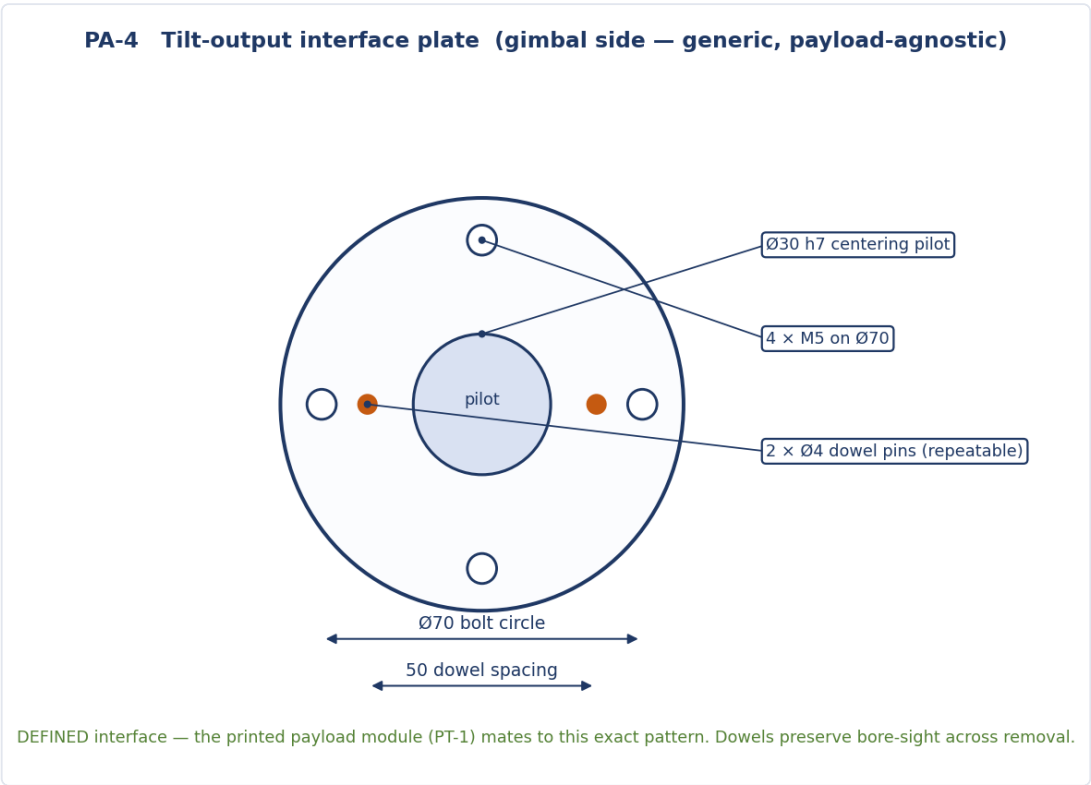
Mount the yoke to the pan turntable, width **W**, square to the pan axis. M5/M6 per schedule, dowel-pinned if possible. **QC:** yoke faces parallel; tilt-bearing bores will sit co-axial (check in Step 4).

Step 4 — Tilt shaft, bearings & motor



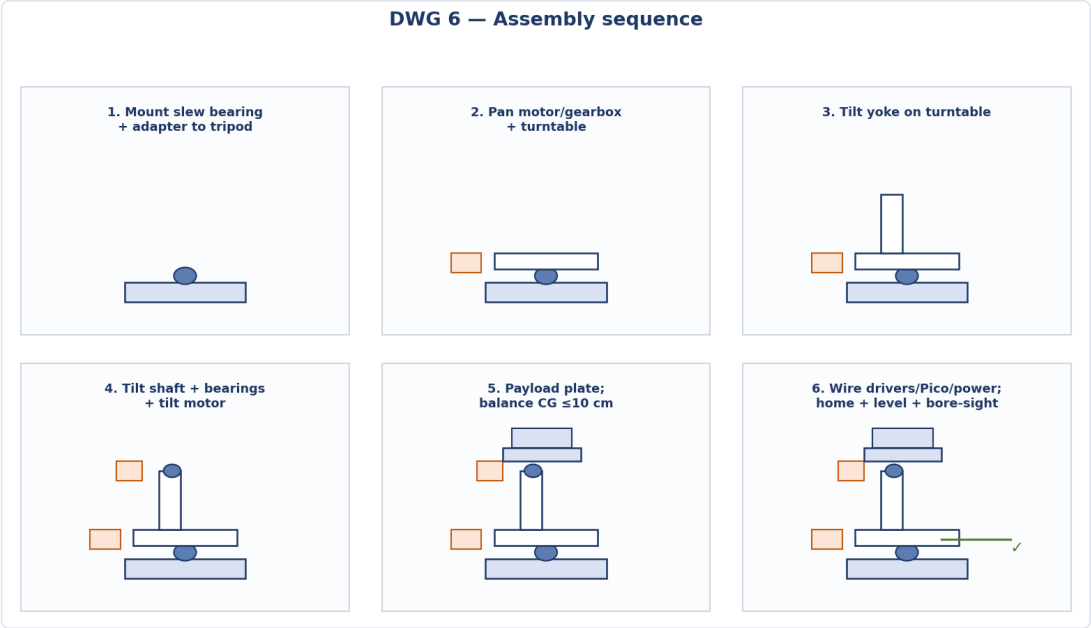
Install the two tilt bearings in the yoke (span **D**), slide the shaft (**Ø E**) through, and set axial position with collars. Mount the tilt motor + gearbox to one yoke wall and couple to the shaft. **QC:** shaft runout ≤ 0.05 mm TIR; tilt axis perpendicular to pan axis $\leq 0.5^\circ$; no bind through full tilt travel.

Step 5 — Tilt-output interface plate



Clamp the **generic interface plate** to the tilt shaft (holes **H**). This is the gimbal's payload mount and is **payload-agnostic** — a Ø70 bolt circle (4×M5), a Ø30 h7 centering pilot, and 2×Ø4 dowel pins. The printed payload module bolts to this exact pattern; the dowels make it repeatable so the payload can be removed and reinstalled without losing bore-sight. **QC:** plate face square to the tilt axis ($\leq 0.5^\circ$); pilot concentric to the shaft; dowels located. The payload itself is built and balanced per `Payload_Top_Module.pdf`.

Step 6 — Alignment, backlash & balance



Final alignment and acceptance — see §7.

7. Acceptance / QC checks

Check	Method	Target
Pan axis $\perp$ base	Square / angle gauge	$\leq 0.5^\circ$
Tilt axis $\perp$ pan axis	Square / angle gauge	$\leq 0.5^\circ$
Shaft / turntable runout	Dial indicator (TIR)	≤ 0.05 mm
Output backlash (each axis)	Indicator at output, reverse load	$\leq 0.1^\circ$ (planetary)
Travel	Manual sweep, full range	No bind; smooth, even torque
Balance	Power off (worm) or holding current	CG ≤ 10 cm from tilt axis; holds any elevation

8. Notes for the fabricator

- Treat all frame members and bearing/shaft fits as the structural priority — recoil + ~10 lb payload load the joints dynamically. Use dowel pins on critical alignments, not bolts alone.
- Bearing fits: shaft to inner race typically a slip fit (g6/h6); housing to outer race a light interference (H7). Confirm against the chosen bearings.

- If using belt reduction, add a tensioner/idler and set tension per the belt spec; if planetary, mount concentric to the axis and check output backlash.
- Keep the design serviceable: the interface plate, motors, and tripod adapter should be removable without disassembling the whole stack. The payload module separates at the PA-4 interface.

9. Dimension schedule

To be finalized from the part scans and final frame sizing. Standard interface dimensions (NEMA 23 pattern, fastener sizes) are fixed in the drawings above.

Ref	Dimension	Value (to confirm)
A	Pan plate width	TBD from frame
B	Pan stack height	TBD from frame
C	Pan bearing bore	TBD from bearing (shaft/coupling through = Ø14)
D	Tilt bearing span	TBD from payload width
E	Tilt drive shaft	Ø14 (gearbox output); far-side trunnion TBD
W	Tilt yoke width	TBD from payload
H	Interface-plate hub bore	Ø14 (on gearbox output shaft)

Interface spec — defined (not TBD): plate Ø90 · bolt circle Ø70 with 4×M5 · centering pilot Ø30 h7 · 2×Ø4 dowel pins at ±25 mm. The payload module mates to this exact pattern. (Payload plate length/width move to the payload-module spec, where they are scan-driven.)

Gearbox-output mounting — defined (PA-6), same on both axes: Ø40 H7 pilot bore · 4×M5 tapped @ 47.14 · Ø16 shaft/coupling clearance bore. Each drive unit (motor + gearbox) ≈ **205 mm** long (motor 56.4 × 138 + gearbox 57 × 67). | J | Tripod bolt circle | TBD from tripod |

Revision A, pre-scan. Re-issue with the schedule populated and component hole patterns dimensioned once the marker, payload, and frame are measured/scanned.